

Functions

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Arrow Functions (ES6)

Arrow function expressions were introduced in ES6. These expressions are clean and concise. The syntax for an arrow function expression does not require the `function` keyword and uses a fat arrow `=>` to separate the parameter(s) from the body.

There are several variations of arrow functions:

- Arrow functions with a single parameter do not require `()` around the parameter list.
- Arrow functions with a single expression can use the concise function body which returns the result of the expression without the `return` keyword.

```
// Arrow function with two parameters
const sum = (firstParam, secondParam) => {
  return firstParam + secondParam;
};
console.log(sum(2,5)); // Prints: 7

// Arrow function with no parameters
const printHello = () => {
  console.log('hello');
};
printHello(); // Prints: hello

// Arrow functions with a single parameter
const checkWeight = weight => {
  console.log(`Baggage weight : ${weight} kilograms.`);
};
checkWeight(25); // Prints: Baggage weight : 25 kilograms.

// Concise arrow functions
const multiply = (a, b) => a * b;
console.log(multiply(2, 30)); // Prints: 60
```

Functions

Functions are one of the fundamental building blocks in JavaScript. A *function* is a reusable set of statements to perform a task or calculate a value. Functions can be passed one or more values and can return a value at the end of their execution. In order to use a function, you must define it somewhere in the scope where you wish to call it.

The example code provided contains a function that takes in 2 values and returns the sum of those numbers.

```
// Defining the function:
function sum(num1, num2) {
  return num1 + num2;
}

// Calling the function:
sum(3, 6); // 9
```

Anonymous Functions

Anonymous functions in JavaScript do not have a name property. They can be defined using the `function` keyword, or as an arrow function. See the code example for the difference between a named function and an anonymous function.

```
// Named function
function rocketToMars() {
  return 'BOOM!';
}

// Anonymous function
const rocketToMars = function() {
  return 'BOOM!';
}
```

Function Expressions

Function *expressions* create functions inside an expression instead of as a function declaration. They can be anonymous and/or assigned to a variable.

```
const dog = function() {  
  return 'Woof!';  
}
```

Function Parameters

Inputs to functions are known as *parameters* when a function is declared or defined. Parameters are used as variables inside the function body. When the function is called, these parameters will have the value of whatever is *passed* in as arguments. It is possible to define a function without parameters.

```
// The parameter is name  
function sayHello(name) {  
  return `Hello, ${name}!`;  
}
```

return Keyword

Functions return (pass back) values using the `return` keyword. `return` ends function execution and returns the specified value to the location where it was called. A common mistake is to forget the `return` keyword, in which case the function will return `undefined` by default.

```
// With return  
function sum(num1, num2) {  
  return num1 + num2;  
}  
  
// Without return, so the function doesn't output the sum  
function sum(num1, num2) {  
  num1 + num2;  
}
```

Function Declaration

Function *declarations* are used to create named functions. These functions can be called using their declared name. Function declarations are built from:

- The `function` keyword.
- The function name.
- An optional list of parameters separated by commas enclosed by a set of parentheses `()`.
- A function body enclosed in a set of curly braces `{}`.

```
function add(num1, num2) {  
  return num1 + num2;  
}
```

Calling Functions

Functions can be *called*, or executed, elsewhere in code using parentheses following the function name. When a function is called, the code inside its function body runs. *Arguments* are values passed into a function when it is called.

```
// Defining the function  
function sum(num1, num2) {  
  return num1 + num2;  
}  
  
// Calling the function  
sum(2, 4); // 6
```